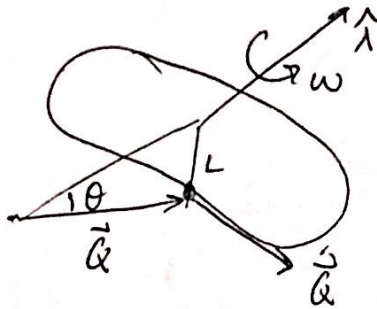


Recall:

- If you had a vector $\vec{Q}$ fixed in object/body/frame B and you have constant axis $\hat{\lambda}$ and rate of rotation ω



$$\dot{\vec{Q}} = \|\vec{Q}\| \sin \theta \cdot \omega \hat{\lambda}$$

$$\dot{\vec{Q}} = \underbrace{\vec{\omega}}_{\omega \hat{\lambda}} \times \vec{Q}$$

Any vector fixed in frame B has $\dot{\vec{Q}} = \vec{\omega}_{B/F} \times \vec{Q}$

Ex) $\hat{i}', \hat{e}_i', \vec{r}_{AB}$
pt. fixed in B

What if $\vec{Q} \neq \text{constant}$? (point not fixed in B)

$$Q_i \hat{e}_i = Q'_i \hat{e}'_i$$

$$\frac{d}{dt} (Q_i \hat{e}_i) = \frac{d}{dt} (Q'_i \hat{e}'_i)$$

$$\dot{Q}_i \hat{e}_i = \dot{Q}'_i \hat{e}'_i + Q'_i \dot{\hat{e}}'_i$$

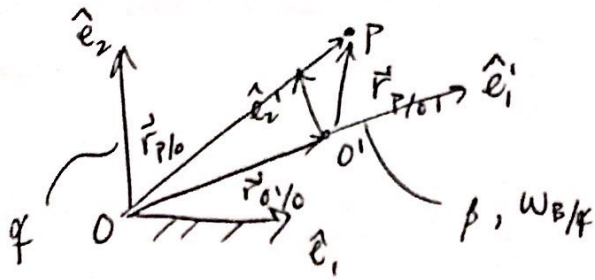
What is $\dot{\hat{e}}'_i$?

$$\hat{e}'_i = \vec{\omega}_{B/F} \times \hat{e}'_i \rightarrow \dot{\vec{Q}} = \underbrace{\dot{Q}'_i \hat{e}'_i}_{\vec{Q}^B} + Q'_i (\vec{\omega}_{B/F} \times \hat{e}'_i)$$

$$\dot{\vec{Q}} = \vec{Q}^B + \vec{\omega}_{B/F} \times \vec{Q}$$

This holds for any 2 frames

Position, velocity, acceleration for point on rigid object



We know:

- Fixed: $\vec{r}_{O'/O}$, $\vec{v}_{O'/F}$, $\vec{a}_{O'/F}$, $\omega_{B/F}$, $\vec{\omega}_{B/F}$
- Moving: $\vec{r}_{P/O'}$, $\vec{v}_{P/B}$, $\vec{a}_{P/B}$
- Find: $\vec{r}_{P/O}$, $\vec{v}_{P/F}$, $\vec{a}_{P/F}$

$$\vec{r}_{P/O} = \vec{r}_{O'/O} + \vec{r}_{P/O'}$$

$$\vec{v}_{P/F} = \frac{d}{dt}(\vec{r}_{P/O}) = \vec{v}_{O'/F} + \frac{d}{dt}(\vec{r}_{P/O'}) = \vec{v}_{O'/F} + \dot{\vec{r}}_{P/O'} + \omega_{B/F} \times \vec{r}_{P/O}$$

$$\vec{v}_{P/F} = \vec{v}_{O'/F} + \vec{\omega}_{B/F} \times \vec{r}_{P/O'} + \vec{v}_{P/B}$$

motion of origin

motion of fixed point in B relative to O'

moving frame velocity

5 term acceleration

$$\vec{a}_{P/F} = \frac{d}{dt}(\vec{v}_{P/O}) = \frac{d}{dt}(\vec{v}_{O'/F} + (\vec{\omega}_{B/F} \times \vec{r}_{P/O'}) + \vec{v}_{P/B})$$

Aside:

$$\dot{\vec{\omega}}_{B/F} = \vec{\omega}_{B/F} + \vec{\omega}_{B/F} \times \vec{\omega}_{B/F}$$

$$\vec{a}_{P/F} = (\vec{a}_{O'/F}) + (\vec{\omega}_{B/F} \times \vec{r}_{P/O'}) + (\vec{\omega}_{B/F} \times (\vec{v}_{P/B} + \vec{\omega}_{B/F} \times \vec{r}_{P/O'})) + (\vec{a}_{P/B} + \vec{\omega}_{P/F} \times \vec{v}_{P/B}) \quad 29.2$$

this term shows
up twice

$$\vec{a}_{P/F} = \vec{a}_{O'/F} + \vec{\omega} \times (\vec{\omega} \times \vec{r}_{P/O'}) + \vec{\omega} \times \vec{r}_{P/O'} + \vec{a}_{P/B} + 2\vec{\omega} \times \vec{v}_{P/B}$$

fixed point in
rotating reference
frame

moving point
in rotating frame